

A Multi-Domain Cyber-Deception Bayesian Game for Resilient IoT Systems

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Abstract—Internet of Things (IoT) infrastructures are increasingly targeted by Advanced Persistent Threats (APTs) that progress methodically across the perception, network, and application layers. Cyber-deception, which strategically deploys decoys (honeypots) to mislead attackers, has emerged as a proactive defense paradigm. However, existing game-theoretic deception models suffer from three structural limitations: they treat each IoT layer in isolation, they assume zero-sum interaction, and they ignore the resilience requirements that govern critical IoT services. We introduce the Multi-Domain Cyber-Deception Bayesian Game (MD-CDBG), a stochastic Bayesian game with incomplete information that couples the three IoT layers under formal resilience and semantic-coherence constraints. We define a novel solution concept, the Resilience-constrained Bayesian Nash Equilibrium (R-BNE), which embeds a hard resilience constraint at every time step, and we solve it with a Bayesian Double Oracle algorithm whose inner oracle uses a Linear Complementarity Program (LCP) to correctly handle the non-zero-sum structure of the game. Through a Lagrangian decomposition, the three coupled sub-games are solved in parallel while enforcing the budget and coherence constraints, and a fallback isolation policy is triggered whenever resilience approaches its threshold. Extensive simulations against three baselines, including the recent GameSec 2025 double-oracle approach, across three attacker profiles (opportunistic, persistent, and state-sponsored) show that the MD-CDBG reduces resilience-constraint violations by 65–72% while inducing rational attacker behavior, demonstrating that modeling deception, belief, and resilience jointly is essential for protecting multi-layer IoT systems.

Index Terms—IoT security, cyber-deception, Bayesian game, Nash equilibrium, resilience, double oracle, linear complementarity, honeypot, advanced persistent threat.

I. INTRODUCTION

The proliferation of Internet of Things (IoT) devices in industrial, medical, and urban infrastructures has dramatically expanded the attack surface available to sophisticated adversaries. Billions of resource-constrained sensors, gateways, and actuators now mediate critical services, and their heterogeneity and scale make conventional perimeter defenses insufficient [1]. Among the threats they face, Advanced Persistent Threats (APTs) are particularly dangerous: they operate methodically over long periods, escalating privileges layer by layer through the canonical three-tier IoT architecture, perception (sensors), network (gateways and routers), and application (cloud services and decision engines) [2].

Traditional intrusion-detection systems are inherently reactive: they detect attacks only after they manifest, by which point a persistent adversary may have already compromised critical assets. Cyber-deception offers a proactive alternative. By deploying decoys (honeypots) that mimic real assets, a defender can mislead the attacker, waste its resources, and reveal its intentions before genuine damage occurs [3], [4]. The core strength of cyber-deception is its proactive nature, transforming static defenses into dynamic, unpredictable, and more resilient security stances [5]. To be effective, however, deception must be optimized through rigorous decision-making under uncertainty, for which game theory provides a principled foundation [6].

Despite a rich literature, existing game-theoretic deception models exhibit three structural gaps when applied to IoT. First, they typically model a single domain or layer, ignoring the cross-layer coupling that characterizes APT progression in IoT; a honeypot strategy optimal for the network layer may be incoherent with the perception or application layer, making decoys easy to unmask. Second, many adopt a zero-sum formulation in which the attacker’s gain is exactly the defender’s loss; this assumption fails for IoT, where energy consumption, deployment cost, and resilience introduce defender objectives that are not mirrored by the attacker. Third, and most critically, prior work largely ignores resilience: the capacity of the system to maintain service continuity under attack, which is the paramount concern for critical IoT deployments [7].

Contributions. This paper addresses these gaps with the following contributions:

- We formulate the Multi-Domain Cyber-Deception Bayesian Game (MD-CDBG), a stochastic Bayesian game coupling the three IoT layers under formal resilience, semantic-coherence, and budget constraints (Section III).
- We define the Resilience-constrained Bayesian Nash Equilibrium (R-BNE), a solution concept that embeds a hard resilience constraint $\Omega(t) \geq \Omega_{\min}$ at every time step (Section V).
- We design R-BNE-DO, a Bayesian Double Oracle algorithm whose inner oracle solves the non-zero-sum game via a Linear Complementarity Program, coordinated by a Lagrangian decomposition over the three layers, and we

prove its convergence (Section VI).

- We evaluate the MD-CDBG against three baselines, including the recent GameSec 2025 double-oracle deception model [5], across three attacker profiles, demonstrating a 65–72% reduction in resilience-constraint violations across all scenarios and substantially more rational attacker behavior, at a defender utility that matches or exceeds the strongest baseline (Section VII).

The remainder of the paper is organized as follows. Section II reviews related work and positions our contribution. Section III formalizes the MD-CDBG. Section IV defines the utility functions. Section V introduces the resilience function and the R-BNE. Section VI presents the R-BNE-DO algorithm and its convergence analysis. Section VII reports the experimental evaluation. Section VIII concludes.

II. RELATED WORK

A. Game-Theoretic Cyber-Deception

Signaling games and Stackelberg games have been widely used to model the strategic interaction between a defender deploying decoys and an attacker probing a network [8]. Pawlick *et al.* [1] provide a comprehensive taxonomy of defensive deception, categorizing approaches by information structure and game type, and book-length treatments establish the scientific foundations of cyber deception and of game theory and machine learning for security [9], [10]. Huang and Zhu [6] model adaptive honeypot engagement as a semi-Markov decision process solved by reinforcement learning. Two-layer signaling games have been proposed to defend cyber-physical systems against attacks that traverse both a cyber and a physical plane [11]. These models capture information asymmetry but generally treat a single domain and assume a static type for the adversary, leaving cross-layer coupling and resilience unaddressed.

B. Double Oracle Methods for Large Games

The Double Oracle (DO) meta-algorithm [12] computes equilibria in large or infinite action spaces by iteratively expanding restricted strategy sets, invoking a best-response oracle for each player until no profitable deviation exists. DO and its variants have proven effective for honeypot-allocation and attack-graph games [13]–[15]. Most relevant to our work, Asghar *et al.* [5] model a dual-domain attack on cyber-physical sensors as a non-zero-sum game solved by double oracle with a greedy, submodular defender oracle, extending their earlier zero-sum multi-domain deception formulation [16]. Crucially, although their game is not zero-sum, they prove it is strategically equivalent to a zero-sum game (in the sense of Moulin and Vial), which lets them solve each restricted game with a single linear program and avoid the linear complementarity problem (LCP) altogether. This equivalence, however, is a property of their particular payoff structure: it does not survive once the defender’s objective carries resilience, energy-budget, and cross-layer coherence penalties that the attacker does not share, as in resilient IoT. Their model also provides no resilience guarantee. We show in Section III that under these

penalties the game is not reducible to zero-sum, so the LCP becomes necessary rather than optional.

C. Bayesian Games and Belief Modeling

Games with incomplete information, introduced by Harsanyi [17], model uncertainty about an opponent’s type through a prior belief updated by Bayes’ rule. In cybersecurity, the attacker’s type (e.g., its sophistication or objective) is rarely known, motivating Bayesian formulations. Wan *et al.* [2] extend this idea with hypergame theory, allowing players to hold subjective and possibly inconsistent views of the same game, and apply it to defensive deception against APTs. However, these approaches maintain a belief over the attacker’s type without coupling that belief to hard operational constraints such as resilience or a global deployment budget.

D. Deception in IoT and Wireless Sensor Networks

A growing body of work targets deception in resource-constrained IoT and wireless sensor networks (WSNs). Anwar *et al.* [3] develop a game-theoretic framework for dynamic cyber-deception in the Internet of Battlefield Things. Sihomnou *et al.* [18], [19] study Bayesian and signaling games to defend IoT devices against energy-depletion and battery-drain denial-of-service attacks, emphasizing the lightweight constraints intrinsic to IoT. He *et al.* [7] propose a deep reinforcement learning approach to time-delay differential game deception for real-time resource deployment. These works recognize IoT-specific constraints but consider a single layer and do not formalize resilience as an equilibrium constraint.

E. Learning-Based and Detection Approaches

Complementary to game theory, machine-learning detectors have been applied to multi-domain intrusion detection, including deep autoencoder models [20] and graph neural networks that deceive network reconnaissance [21]. Sayed *et al.* [4] combine game theory with learning to defend against zero-day attacks. Such detectors learn attacker behavior reactively but lack the forward-looking equilibrium reasoning and resilience guarantees that a game-theoretic formulation provides; integrating such a learned detector as an additional oracle within our framework is a promising direction (Section VIII).

F. Positioning

To our knowledge, the MD-CDBG is the first model to (i) couple all three IoT layers in a single Bayesian deception game, (ii) preserve the non-zero-sum structure through an LCP-based oracle rather than a zero-sum proxy, and (iii) embed resilience as a hard constraint within the equilibrium concept itself. Table I summarizes how the MD-CDBG differs from representative prior work.

III. SYSTEM MODEL: MD-CDBG

A. Game Definition

We model the defender–attacker interaction across the three IoT layers as a multi-domain stochastic Bayesian game:

$$\mathcal{G} = \langle \mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3, \mathcal{C}, \Omega \rangle, \quad (1)$$

TABLE I
POSITIONING OF THE MD-CDBG AGAINST REPRESENTATIVE PRIOR WORK.

Work	Multi-layer	Non-zero-sum	Bayesian belief	Resil. constr.
Pawlick [1]	✗	✓	✓	✗
Huang [6]	✗	✓	✗	✗
Wan [2]	✗	✓	✓	✗
Anwar [3]	✗	✓	✗	✗
Asghar [5]	✓	✗	✗	✗
Nguemkam [13]	✗	✓	✓	✗
MD-CDBG (ours)	✓	✓	✓	✓

The “non-zero-sum” column indicates whether the equilibrium is computed on the genuine non-zero-sum game. Asghar [5] formulate a non-zero-sum game but reduce it to a strategically equivalent zero-sum game; MD-CDBG instead solves the non-zero-sum game directly via the LCP.

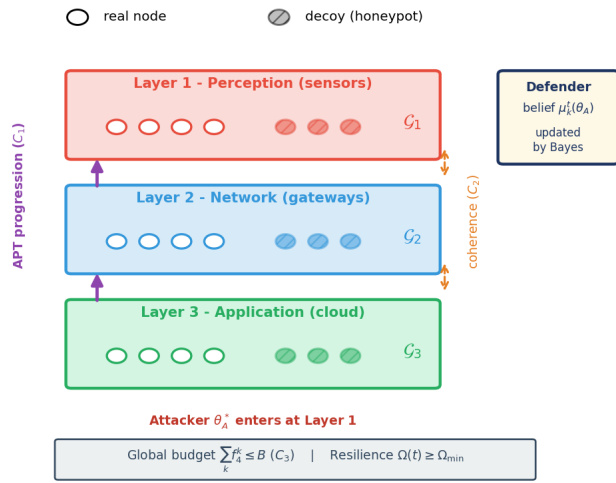


Fig. 1. Architecture of the MD-CDBG. The attacker progresses upward through the three layers (C_1); decoys at adjacent layers must remain coherent (C_2); the defender maintains a per-layer Bayesian belief $\mu_k^t(\theta_A)$ over the attacker type, under a global budget (C_3) and a resilience constraint $\Omega(t) \geq \Omega_{min}$.

where $\mathcal{G}_k$ is the sub-game at layer $k \in \{1, 2, 3\}$ (perception, network, application), $\mathcal{C} = \{C_1, C_2, C_3\}$ is the set of coupling constraints linking the sub-games, and Ω is the system resilience function.

Each sub-game is a tuple

$$\mathcal{G}_k = \langle \mathcal{N}_k, \hat{\mathcal{N}}_k, \mathcal{A}_k^D, \mathcal{A}_k^A, \Theta_A, u_k^D, u_k^A \rangle, \quad (2)$$

comprising $|\mathcal{N}_k|$ real nodes and $|\hat{\mathcal{N}}_k|$ decoy nodes, defender and attacker action sets $\mathcal{A}_k^D, \mathcal{A}_k^A$, a set of attacker types $\Theta_A = \{\theta_{opp}, \theta_{pers}, \theta_{state}\}$ corresponding to opportunistic, persistent, and state-sponsored adversaries, and layer-level utilities u_k^D, u_k^A . The defender chooses a mixed strategy $\sigma_k^D \in \Delta(\mathcal{A}_k^D)$; the attacker of type θ_A chooses $\sigma_k^A(\cdot | \theta_A) \in \Delta(\mathcal{A}_k^A)$. Figure 1 illustrates the overall architecture, and Table II summarizes the main notation.

TABLE II
MAIN NOTATION.

Symbol	Meaning
$\mathcal{G}_k$	Sub-game at layer k (perception/network/app.)
$\mathcal{N}_k, \hat{\mathcal{N}}_k$	Real and decoy nodes at layer k
Θ_A	Attacker types (opp./pers./state)
$\mathcal{A}_k^D, \mathcal{A}_k^A$	Defender / attacker action sets at layer k
$\Delta(\mathcal{A}_k^D), \Delta(\mathcal{A}_k^A)$	Probability simplices over actions
$a_{k,i}^D, a_{k,i}^A$	i -th defender / attacker action at layer k
$a^A(t), a^D(t)$	Joint attacker / defender action profiles
$a_k^A(t)$	Observed attacker action at layer k , time t
$\mathbf{s}, \mathbf{s}_k(t)$	Global / per-layer system state profile
$\mathcal{T}$	State-transition function
θ_k^*	True (realized) attacker type
σ_k^D, σ_k^A	Defender / attacker mixed strategies
$\mu_k^t(\theta_A)$	Bayesian belief on type at layer k , time t
L_t	Likelihood-ratio process
$\delta_k(t)$	Compromise ratio of layer k
f_1^k	Deception effectiveness
f_3^k, f_4^k	Energy overhead and deployment cost
q_k	Composite decoy quality
$q_k^{\text{func}}, q_k^{\text{stat}}, q_k^{\text{temp}}$	Functional / statistical / temporal realism
$r_{k,i}^{\text{func}}$	Intrinsic realism of action $a_{k,i}^D$
ω_i	Decoy-quality component weights
$d_k^{\text{eff}}, d_k^{\text{vuln}}$	Detection capability (active/passive)
$\psi_k(\theta_A)$	Type-specific detection efficiency
κ	Passive-detection coefficient
Coh_k	Semantic coherence between layers $k, k+1$
Ω_k, Ω	Local / global resilience
$\text{res}_k, \text{abs}_k, \text{rec}_k$	Resistance / absorption / recovery terms
w_1, w_2, w_3	Adaptive resilience-component weights
ρ_k	Resistance threshold (layer k)
τ_k, T_k^{max}	Per-node and max recovery time
$\Omega_{\min}$	Resilience threshold (hard constraint)
Π_{Ω}^D	Admissible (feasible) defender policy set
σ^{D*}, σ^{A*}	R-BNE defender / attacker policies
g_k, c_k, ρ_k^A	Attacker gain / cost / detection risk
$v_{k,i}^A, p_{k,i}^A$	Asset value, success probability
$r_{k,i}^A, \lambda_{k,i}$	Action resource cost, detection sensitivity
$e_{k,i}^D, c_{k,i}^D$	Defender unit energy / deployment cost
u_k^D, u_k^A	Local defender / attacker utility
U^D, U^A	Global defender / attacker utility
$\mathcal{L}_k$	Per-layer Lagrangian
M_k^D, M_k^A	Defender / attacker payoff matrices (Bayes-DO)
R_D, R_A	Restricted action sets (double oracle)
α_i, γ_i	Defender / attacker utility weights
$\beta_k, \beta_k^A, \beta_k^\Omega$	Layer aggregation weights
γ	Discount factor; T : time horizon
η_t, η_0	Gradient step size (Robbins–Monro)
ϵ, ε	Stability constant; convergence tolerance
λ	Lagrange multiplier (budget C_3)
ν_k	Lagrange multiplier (coherence C_2)
$\delta_{\min}, B$	Coherence threshold / budget

B. Coupling Constraints

The three sub-games are linked by three constraints that make the MD-CDBG a genuinely multi-domain game rather than three independent games:

- (C_1) APT progression: layer k is accessible only if at least one node in layer $k-1$ is compromised. This enforces the

sequential escalation characteristic of APTs: an attacker must breach the perception layer before the network layer, and the network before the application.

- (C_2) Semantic coherence: decoys at adjacent layers must be mutually consistent, $\text{Coh}(\hat{\mathcal{N}}_k, \hat{\mathcal{N}}_{k+1}) \geq \delta_{\min}$. An attacker that observes a network decoy inconsistent with the perception decoys it has already seen can unmask the deception.
- (C_3) Global budget: the total deployment cost across all layers must not exceed a budget B , i.e. $\sum_k f_4^k \leq B$, reflecting the finite operational resources of an IoT operator.

C. State and Transition

The system state $\mathbf{s}(t) = (\mathbf{s}_1(t), \mathbf{s}_2(t), \mathbf{s}_3(t))$ records, for each layer, the set of compromised nodes. The compromise ratio of layer k is $\delta_k(t) = |\text{compromised}_k|/|\mathcal{N}_k|$. At each step, the realized defender and attacker actions induce a stochastic transition $\mathbf{s}(t+1) = \mathcal{T}(\mathbf{s}(t), a^A(t), a^D(t))$: an attacker action targeting a real node succeeds with a technical success probability $p_{k,i}^A$, unless it is trapped by a decoy (probability f_1^k), while a defender isolation action can recover a compromised node.

D. Bayesian Belief

Since the attacker's true type θ_A^* is unobservable, the defender maintains a belief $\mu_k^t(\theta_A) \in \Delta(\Theta_A)$ per layer k and time t , initialized at the prior $\mu_k^0(\theta_A) = p(\theta_A)$ and updated after each observed action $a_k^A(t)$ via Bayes' rule:

$$\mu_k^{t+1}(\theta_A) = \frac{\sigma_k^A(a_k^A(t) | \theta_A) \mu_k^t(\theta_A)}{\sum_{\theta' \in \Theta_A} \sigma_k^A(a_k^A(t) | \theta') \mu_k^t(\theta')}. \quad (3)$$

Lemma 1 (Belief convergence). *Suppose the type-conditional strategies $\{\sigma_k^A(\cdot | \theta_A)\}_{\theta_A \in \Theta_A}$ are pairwise distinct (identifiability) and layer k is visited infinitely often. Then, under the true type θ_A^* , the belief $\mu_k^t(\theta_A^*) \rightarrow 1$ almost surely as $t \rightarrow \infty$; that is, the belief concentrates on the true type. The result follows because, for every $\theta \neq \theta_A^*$, the likelihood-ratio process $L_t(\theta) = \mu_k^t(\theta)/\mu_k^t(\theta_A^*)$ is a non-negative martingale under θ_A^* and hence converges almost surely; the strict positivity of the Kullback–Leibler divergence between distinct type-conditional distributions forces its limit to 0, so all false types vanish and $\mu_k^t(\theta_A^*) \rightarrow 1$.*

IV. UTILITY FUNCTIONS

A. Decoy Quality and Detection

The realism of a layer- k decoy deployment is captured by a composite quality $q_k(\sigma_k^D) \in [0, 1]$ combining functional, statistical, and temporal realism:

$$q_k(\sigma_k^D) = \omega_1 q_k^{\text{func}} + \omega_2 q_k^{\text{stat}} + \omega_3 q_k^{\text{temp}}, \quad \sum_i \omega_i = 1, \quad (4)$$

where the functional realism $q_k^{\text{func}}(\sigma_k^D) = \sum_i \sigma_k^D(a_{k,i}^D) r_{k,i}^{\text{func}}$ is the strategy-weighted realism of the chosen decoy actions ($r_{k,i}^{\text{func}} \in [0, 1]$ being the intrinsic realism of action $a_{k,i}^D$), and the statistical and temporal components measure how closely decoy traffic matches that of real nodes.

The attacker's detection capability combines an active and a passive component through a complementary-probability law:

$$d_k(\sigma_k^A, \sigma_k^D) = 1 - (1 - d_k^{\text{eff}})(1 - d_k^{\text{vuln}}), \quad (5)$$

where the active component scales the attacker's probing investment by its type-specific efficiency,

$$d_k^{\text{eff}}(\sigma_k^A) = \sigma_k^A(a_k^{\text{probe}}) \cdot \psi_k(\theta_A), \quad (6)$$

and the passive component reflects the intrinsic detectability of imperfect decoys, $d_k^{\text{vuln}}(\sigma_k^D) = \kappa(1 - q_k)$ with $\kappa \in [0, 1]$. Equation (5) expresses that a decoy escapes detection only if it survives both the active probe and the passive inspection, hence the product of the two survival probabilities.

B. Deception Effectiveness

The deception effectiveness at layer k multiplies the decoy ratio, the decoy quality, and the complement of the detection capability:

$$f_1^k(\sigma_k^D, \sigma_k^A) = \frac{|\hat{\mathcal{N}}_k|}{|\mathcal{N}_k| + |\hat{\mathcal{N}}_k|} \cdot q_k(\sigma_k^D) \cdot (1 - d_k(\sigma_k^A, \sigma_k^D)). \quad (7)$$

The three factors admit a clear reading: more decoys raise the probability that the attacker targets one; higher quality reduces the chance the attacker recognizes it; and lower detection capability increases the chance the decoy goes unnoticed.

C. Semantic Coherence

The coherence constraint C_2 penalizes abrupt quality discontinuities between adjacent layers, which an attacker can exploit to unmask the deception. We measure coherence by the normalized similarity of the decoy quality at two adjacent layers:

$$\text{Coh}_k(\sigma_k^D, \sigma_{k+1}^D) = 1 - \frac{|q_k(\sigma_k^D) - q_{k+1}(\sigma_{k+1}^D)|}{\max(q_k, q_{k+1}, \epsilon)}, \quad (8)$$

where $\epsilon > 0$ avoids division by zero. $\text{Coh}_k \in [0, 1]$ equals 1 when the two layers deploy decoys of identical quality and decreases as their realism diverges; constraint C_2 requires $\text{Coh}_k \geq \delta_{\min}$ for all adjacent pairs.

D. Defender Utility

Definition 1 (Defender utility). *The defender's utility at layer k is a weighted linear combination of four partially conflicting objectives:*

$$u_k^D(\sigma_k^D, \sigma_k^A(\cdot | \theta_A)) = \underbrace{\alpha_1 f_1^k}_{\text{deception}} + \underbrace{\alpha_2 \Omega_k(\sigma_k^D)}_{\text{resilience}} - \underbrace{\alpha_3 f_3^k(\sigma_k^D)}_{\text{energy}} - \underbrace{\alpha_4 f_4^k(\sigma_k^D)}_{\text{cost}}, \quad (9)$$

with $\sum_i \alpha_i = 1$, $\alpha_i \geq 0$. The energy overhead and operational deployment cost are the strategy-weighted sums of the per-action unit costs:

$$f_3^k(\sigma_k^D) = \sum_i \sigma_k^D(a_{k,i}^D) e_{k,i}^D, \quad f_4^k(\sigma_k^D) = \sum_i \sigma_k^D(a_{k,i}^D) c_{k,i}^D, \quad (10)$$

where $c_{k,i}^D$ and $c_{k,i}^D$ are the unit energy and deployment costs of defender action $a_{k,i}^D$ (both normalized to $[0, 1]$).

Because the true type is unobservable, the defender evaluates u_k^D in expectation over its current belief, giving the belief-weighted local utility

$$\bar{u}_k^D(\sigma_k^D(\mu_k^t), \mu_k^t) = \mathbb{E}_{\theta_A \sim \mu_k^t} [u_k^D(\sigma_k^D, \sigma_k^A(\cdot | \theta_A))], \quad (11)$$

and the global defender utility aggregates over layers:

$$U^D(\sigma^D(\mu^t), \mu^t) = \sum_{k=1}^3 \beta_k \bar{u}_k^D(\sigma_k^D(\mu_k^t), \mu_k^t), \quad (12)$$

with $\sum_k \beta_k = 1$, $\beta_k > 0$ reflecting the relative criticality of each layer.

E. Attacker Utility

The utility of attacker type θ_A at layer k balances expected asset gain g_k^A , resource cost c_k^A , and detection risk ρ_k^A :

$$u_k^A(\sigma_k^A, \sigma_k^D | \theta_A) = \gamma_1 g_k^A - \gamma_2 c_k^A - \gamma_3 \rho_k^A, \quad (13)$$

with $\sum_i \gamma_i = 1$. The three components are

$$g_k^A = \sum_i \sigma_k^A(a_{k,i}^A) v_{k,i}^A (1 - f_1^k) p_{k,i}^A, \quad (14)$$

$$c_k^A = \sum_i \sigma_k^A(a_{k,i}^A) r_{k,i}^A, \quad (15)$$

$$\rho_k^A = \sum_i \sigma_k^A(a_{k,i}^A) f_1^k \lambda_{k,i}, \quad (16)$$

where $v_{k,i}^A$ is the value of the asset targeted by action $a_{k,i}^A$, $p_{k,i}^A$ its technical success probability, $r_{k,i}^A$ its resource cost, and $\lambda_{k,i}$ its detection sensitivity. Crucially, the deception term f_1^k enters both g_k^A (as a factor $1 - f_1^k$, reducing gain when the attacker hits a decoy) and ρ_k^A (increasing the risk of being trapped). This dual effect forces the attacker to balance aggression against caution, producing genuinely mixed strategies at equilibrium, as illustrated in Figure 2. The global attacker utility aggregates over layers subject to the C_1 accessibility indicator:

$$U^A(\sigma^A, \sigma^D | \theta_A) = \sum_{k=1}^3 \beta_k^A \mathbb{E}_s [u_k^A \cdot \mathbf{1}[\mathcal{A}_k^A(s) \neq \emptyset]]. \quad (17)$$

Remark 1 (Structural asymmetry). *The defender's global utility U^D is an expectation over attacker types (via the belief μ_k^t), whereas the attacker's utility U^A is an expectation over system states (which nodes are decoys). This double incomplete information is the precise source of the game's non-zero-sum structure: the resilience, energy-budget, and coherence terms in U^D have no counterpart in U^A , so $u_k^D \neq -u_k^A$ in general. For the dual-domain game of [5] the defender and attacker payoffs differ only by a column-constant term, so the game is strategically equivalent to a zero-sum game and is solvable by a linear program. Here that property fails: the constraint-driven penalties in U^D are not column-constant, the bimatrix $M_k^D + M_k^A$ is not rank-one, and no zero-sum reduction exists. The restricted game must therefore*

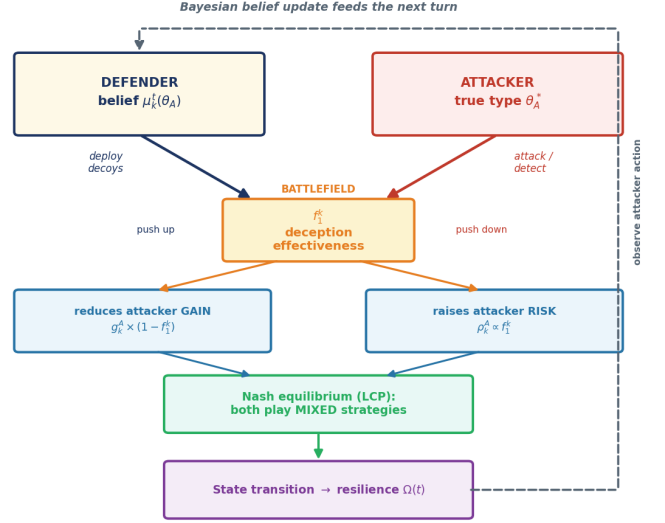


Fig. 2. Per-turn interaction between the two players. Both act simultaneously on the deception effectiveness f_1^k : the defender pushes it up by deploying more and better decoys, the attacker pushes it down through detection effort. Because f_1^k both reduces the attacker's gain and raises its risk, neither player can play a pure strategy, yielding a mixed Nash equilibrium computed by the LCP. The resulting state transition updates resilience $\Omega(t)$, and the observed attacker action feeds the Bayesian belief update for the next turn.

be solved as a genuine non-zero-sum bimatrix game via the LCP (Section VI).

V. RESILIENCE AND THE R-BNE

A. Resilience Function

Following the resistance-absorption-recovery decomposition common in cyber-physical resilience analysis [22]–[24], the local resilience at layer k is a weighted sum of three components:

$$\Omega_k(t) = w_1 \cdot \text{res}_k(t) + w_2 \cdot \text{abs}_k(t) + w_3 \cdot \text{rec}_k(t), \quad (18)$$

defined as

$$\text{res}_k(t) = \mathbf{1}[1 - \delta_k(t) \geq \rho_k], \quad (19)$$

$$\text{abs}_k(t) = \max(0, 1 - \delta_k(t)), \quad (20)$$

$$\text{rec}_k(t) = \mathbf{1}[\delta_k(t) | \mathcal{N}_k | \tau_k \leq T_k^{\max}], \quad (21)$$

where ρ_k is the resistance threshold, τ_k the per-node recovery time, and $T_k^{\max}$ the maximum acceptable recovery time. Resistance indicates whether the uncompromised fraction exceeds ρ_k ; absorption measures the remaining healthy fraction; and recovery indicates whether the compromised nodes can be restored within the acceptable time budget. The weights w_1, w_2, w_3 adapt to the global compromise level $\bar{\delta}(t) = \frac{1}{3} \sum_k \delta_k(t)$: resistance dominates when the system is healthy, absorption when it is partially compromised, and recovery when damage is extensive. The global resilience is the weighted aggregation

$$\Omega(t) = \sum_{k=1}^3 \beta_k^\Omega \Omega_k(t), \quad \sum_k \beta_k^\Omega = 1. \quad (22)$$

B. Solution Concept

Definition 2 (Admissible policy set). *The set Π_Ω^D of admissible defender policies comprises all belief-dependent behavioral policies $\sigma^D : \Delta(\Theta_A) \rightarrow \Delta(\mathcal{A}^D)$ that satisfy the resilience constraint $\Omega(t) \geq \Omega_{\min}(t)$ for all t and the coupling constraints $\mathcal{C}$.*

Definition 3 (R-BNE). *A Resilience-constrained Bayesian Nash Equilibrium is a strategy profile $(\sigma^{D*}(\cdot), \sigma^{A*}(\cdot | \theta_A^*))$ satisfying, for all belief states $\mu^t \in \Delta(\Theta_A)$:*

$$(i) \sigma^{D*}(\mu^t) = \arg \max_{\sigma^D \in \Pi_\Omega^D} \mathbb{E}_{\theta_A \sim \mu^t} [U^D(\sigma^D(\mu^t), \sigma^{A*})], \quad (23)$$

$$(ii) \sigma^{A*}(\cdot | \theta_A^*) = \arg \max_{\sigma^A \in \Delta(\mathcal{A}^A)} U^A(\sigma^{D*}, \sigma^A | \theta_A^*), \quad (24)$$

$$(iii) \Omega(t) \geq \Omega_{\min}(t) \quad \forall t \geq 0. \quad (25)$$

The R-BNE differs from a standard Bayesian Nash equilibrium [25] in three respects. First, the defender strategy $\sigma^{D*}(\mu^t)$ is a belief-dependent behavioral policy, mapping the current belief to a mixed action rather than being a fixed distribution. Second, the feasible set Π_Ω^D imposes a hard resilience constraint at every time step. Third, condition (ii) uses the attacker's true utility U^A , not the defender's constraint-penalized objective, ensuring that the attacker's rationality is modeled independently of the defender's internal penalties.

Proposition 1 (Existence). *If (E1) the admissible set Π_Ω^D is non-empty, (E2) each u_k^D is continuous and concave in σ_k^D , and (E3) the action simplices are compact and convex, then the MD-CDBG admits at least one R-BNE.*

VI. R-BNE-DO ALGORITHM

A. Lagrangian Decomposition

The coupling constraints C_2 and C_3 are dualized via non-negative Lagrange multipliers ν_k (semantic coherence at layer k) and λ (global budget). For fixed multipliers, the Lagrangian separates into three independent layer sub-problems:

$$\mathcal{L} = \sum_{k=1}^3 \mathcal{L}_k(\sigma_k^D, \sigma_k^A, \lambda, \nu_k), \quad (26)$$

where each local Lagrangian augments the belief-weighted defender utility with the budget and coherence penalties:

$$\mathcal{L}_k = \beta_k \mathbb{E}_{\theta_A \sim \mu_k^t} [u_k^D] - \lambda f_4^k - \nu_k \max(0, \delta_{\min} - \text{Coh}_k). \quad (27)$$

This separation is what enables the three sub-games to be solved in parallel. The dual problem $\min_{\lambda, \nu \geq 0} \max_{\sigma^D \in \Pi_\Omega^D, \sigma^A} \mathcal{L}$ is solved by dual gradient ascent: the multipliers are updated in the direction of their respective constraint violations,

$$\lambda^{t+1} = \left[\lambda^t + \eta_t \left(\sum_k f_4^k - B \right) \right]^+, \quad (28)$$

$$\nu_k^{t+1} = \left[\nu_k^t + \eta_t (\delta_{\min} - \text{Coh}_k) \right]^+, \quad (29)$$

with step $\eta_t = \eta_0 / \sqrt{t}$ satisfying the Robbins–Monro conditions $\sum_t \eta_t = \infty$ and $\sum_t \eta_t^2 < \infty$. The multipliers act as

Algorithm 1 Bayes-DO: Bayesian Double Oracle for $\mathcal{G}_k$

Require: $\mathcal{G}_k$, multipliers (λ, ν_k) , belief $\mu_k^t(\theta_A)$

Ensure: Local BNE $(\sigma_k^{D*}, \sigma_k^{A*})$

```

1: Initialize  $R_D \leftarrow \{a_{k,1}^D\}$ ,  $R_A \leftarrow \{a_{k,1}^A\}$ 
2: repeat
3:    $(\tilde{\sigma}_k^D, \tilde{\sigma}_k^A) \leftarrow \text{NE}(M_k^D|_{R_D \times R_A}, M_k^A|_{R_D \times R_A})$  via LCP
4:    $a_k^{D*} \leftarrow \arg \max_a \mathbb{E}_{\theta_A \sim \mu_k^t} [\mathcal{L}_k(a, \tilde{\sigma}_k^A)]$ 
5:    $a_k^{A*} \leftarrow \arg \max_a u_k^A(\tilde{\sigma}_k^D, a | \theta_A^*)$ 
6:   if  $a_k^{D*} \in R_D$  and  $a_k^{A*} \in R_A$  then
7:     return  $(\tilde{\sigma}_k^D, \tilde{\sigma}_k^A)$  {exact equilibrium found}
8:   else
9:      $R_D \leftarrow R_D \cup \{a_k^{D*}\}$ ;  $R_A \leftarrow R_A \cup \{a_k^{A*}\}$ 
10:  end if
11: until convergence

```

adaptive shadow prices: λ rises when the budget is exceeded, ν_k rises when inter-layer coherence falls below $\delta_{\min}$.

B. Bayesian Double Oracle

The inner oracle (Algorithm 1) computes the local Bayesian Nash equilibrium of sub-game $\mathcal{G}_k$ for fixed multipliers. Because the MD-CDBG is non-zero-sum (Remark 1), the restricted-game equilibrium cannot be computed by a single minimax linear program; instead, we use a Linear Complementarity Program [26] with two distinct payoff matrices. Let e_i denote the pure defender action $a_{k,i}^D$ and e_j the pure attacker action $a_{k,j}^A$. The defender matrix M_k^D encodes the belief-weighted Lagrangian payoff, and the attacker matrix M_k^A encodes the attacker's true utility:

$$M_k^D[i, j] = \mathbb{E}_{\theta_A \sim \mu_k^t} [u_k^D(e_i, e_j | \theta_A)] - \lambda f_4^k(e_i) - \nu_k \max(0, \delta_{\min} - \text{Coh}_k(e_i, \sigma_{k+1}^D)), \quad (30)$$

$$M_k^A[i, j] = u_k^A(e_i, e_j | \theta_A^*). \quad (31)$$

Two points distinguish them. First, M_k^D is an expectation over the belief μ_k^t (the defender does not know the type), whereas M_k^A uses the attacker's true type θ_A^* . Second, the budget and coherence penalties (λ, ν_k) appear only in M_k^D : they are the defender's internal constraints and do not affect the attacker's genuine incentives. Consequently $M_k^D[i, j] + M_k^A[i, j] \neq 0$ in general, which is precisely why a non-zero-sum LCP is required rather than a single minimax LP. The defender's best response (Step 2) is computed in expectation over the belief, whereas the attacker's best response (Step 3) uses its true utility, never the Lagrangian.

Remark 2 (Exact termination). *Bayes-DO is a combinatorial algorithm with exact termination: the expansion check at Step 6 returns true precisely when no profitable deviation exists outside the restricted sets, which is the definition of a Nash equilibrium of the full game. No approximation tolerance is required, since equilibria are typically supported on a small subset of actions.*

Algorithm 2 R-BNE-DO: R-BNE via Bayesian Double Oracle

Require: MD-CDBG, B , $\Omega_{\min}$, $\delta_{\min}$, η_0 , ε , T

```

1: Initialize  $\lambda^0 \leftarrow 0$ ,  $\nu_k^0 \leftarrow 0$ ,  $\mu_k^0(\theta_A) \leftarrow p(\theta_A)$ 
2: for  $t = 0, \dots, T-1$  do
3:   [A] Propagate  $C_1$ ; compute  $\Omega(t)$ 
4:   for  $k = 1, 2, 3$  in parallel do
5:     if  $\Omega(t) < \Omega_{\min}$  then
6:       Activate fallback (isolation) policy
7:     else
8:        $(\sigma_k^{D,t}, \sigma_k^{A,t}) \leftarrow \text{Bayes-DO}(\mathcal{G}_k, \lambda^t, \nu_k^t, \mu_k^t)$ 
9:     end if
10:  end for
11:  [B] Sample  $a^A(t)$ ,  $a^D(t)$ ; transition  $s(t+1)$ 
12:  [C] Update belief  $\mu_k^{t+1}$  via (3)
13:  [D] Update  $\lambda^{t+1}, \nu_k^{t+1}$  by dual ascent
14:  if  $\max(\|\Delta\sigma^D\|, |\Delta\lambda|, \max_k |\Delta\nu_k|) < \varepsilon$  then
15:    return  $(\sigma^{D,t+1}, \sigma^{A,t+1})$ 
16:  end if
17: end for

```

C. Main Algorithm

The outer algorithm R-BNE-DO (Algorithm 2) operates over a horizon T in four phases per step. Phase A propagates the C_1 constraint, checks resilience feasibility, and calls Bayes-DO per layer in parallel; if the global resilience would fall below $\Omega_{\min}$, a fallback policy prioritizing node isolation is activated to restore resilience. Phase B samples actions and transitions the state. Phase C updates the per-layer belief. Phase D updates the Lagrange multipliers.

D. Convergence Analysis

Theorem 1 (Convergence). *Under the conditions of Proposition 1, with Robbins–Monro steps and assuming (A1) uniqueness of the restricted Nash equilibrium at each Bayes-DO iteration (or a fixed selection rule), and (A2) local concavity of $\mathcal{L}_k$ in σ_k^D , the R-BNE-DO algorithm converges to an R-BNE.*

Proof sketch. Convergence combines three arguments operating at the three nested levels. (i) Inner level: for fixed multipliers, Bayes-DO terminates in finitely many iterations at the exact restricted Nash equilibrium, computed by the LCP [26]; assumption (A1) prevents cycling. (ii) Outer level: dual gradient ascent with diminishing steps converges to the saddle point of the Lagrangian (26) under Slater’s strong-duality condition, which holds because $\Pi_{\Omega}^D \neq \emptyset$; at convergence the complementary slackness conditions hold, $\lambda^* > 0$ only if $\sum_k f_4^k = B$, and $\nu_k^* > 0$ only if $\text{Coh}_k = \delta_{\min}$. (iii) Belief level: by Lemma 1, $\mu_k^t(\theta_A) \rightarrow 1$ almost surely, so the defender’s best response converges to the one under perfect type knowledge, tightening the bound on the iterations needed. $\square$

Proposition 2 (Per-call complexity). *Let $r_D = |R_D|$ and $r_A = |R_A|$ be the restricted-set sizes at termination. The LCP solved by support enumeration [27] costs $\mathcal{O}(2^{r_D} 2^{r_A} (r_D + r_A)^3)$. In the MD-CDBG, $r_D \leq 5$ and $r_A \leq 6$, so each call costs at*

most $\approx 2.7 \times 10^6$ elementary operations, and the number of outer iterations to reach precision ε is $\mathcal{O}(1/\varepsilon^2)$.

VII. EXPERIMENTAL EVALUATION

A. Implementation

The MD-CDBG and the R-BNE-DO algorithm were implemented from scratch in Python 3 using NumPy for the numerical core and SciPy for linear programming. The custom simulator faithfully mirrors Algorithms 1 and 2: it maintains a per-layer state and belief, runs the four phases (feasibility check, Bayes-DO, transition, belief and multiplier updates) at each of the T time steps, and records the per-step utilities, resilience, and beliefs. The non-zero-sum local equilibria are computed by a dedicated Linear Complementarity Program solver based on support enumeration [27], which returns the exact mixed Nash equilibrium of each restricted bimatrix game; the zero-sum baseline instead calls a minimax linear program through SciPy’s linprog. Each experiment runs on a standard CPU in a few minutes; no GPU is required. To support reproducibility, the complete source code and a ready-to-run notebook are made available.¹

B. Setup

We simulate a three-layer IoT system with $|\mathcal{N}_k| = \{15, 8, 4\}$ real nodes and $|\hat{\mathcal{N}}_k| = \{10, 5, 3\}$ decoys, over a horizon $T = 50$ and 10 independent replications per scenario. The three scenarios correspond to the three attacker types: O (opportunistic, $\psi = 0.20$), P (persistent, $\psi = 0.50$), and E (state-sponsored, $\psi = 0.65$ – 0.70). We set $\Omega_{\min} = 0.55$, $\delta_{\min} = 0.60$, budget $B = 1.2$, discount $\gamma = 0.95$, and step $\eta_0 = 0.10$. Defender utility weights are $\alpha = (0.55, 0.32, 0.07, 0.06)$. Parameter ranges are calibrated from the IoT-security literature: detection efficiencies $\psi_k(\theta_A)$ follow the attacker-sophistication levels reported for APT honeypot detection [2], decoy ratios reflect typical honeypot-deployment densities, and the resilience threshold $\Omega_{\min}$ is set to a service-continuity level consistent with critical-IoT requirements. All results are reported with 95% confidence intervals.

C. Baselines

We compare R-BNE-DO+LCP against three baselines spanning the spectrum from naive to state-of-the-art, chosen so that each isolates one design element:

- No-Deception: the defender performs passive monitoring only and deploys no decoys; the attacker plays a rational best response. This is the lower bound that quantifies the value of deception itself.
- Random: decoys are deployed uniformly at random across actions, with no strategic reasoning. Comparing against it isolates the contribution of optimized placement over mere presence of decoys.
- Asghar-2025 [5]: the state-of-the-art GameSec 2025 double-oracle deception model, ported to our IoT setting: a greedy submodular defender oracle with restricted

¹Source code and Colab notebook available upon request / at the project repository.

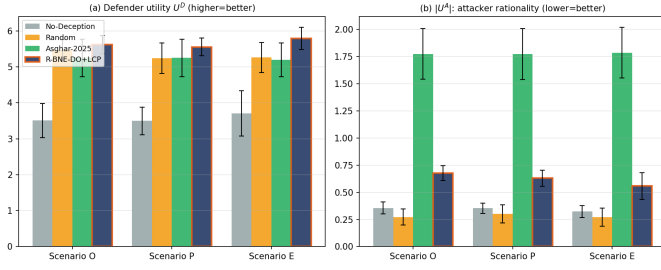


Fig. 3. Defender utility U^D and absolute attacker utility $|U^A|$ across the three scenarios (10 runs, 95% CI). The MD-CDBG induces a far more rational attacker ($|U^A| \approx 0.6$) than the zero-sum proxy of Asghar-2025 ($|U^A| \approx 2.0$).

games solved by a linear program under the authors’ strategic zero-sum equivalence, with no Bayesian belief and no resilience constraint. It shares the double-oracle backbone with our method, so comparing against it isolates the two genuine differences: the non-zero-sum (LCP) treatment and the resilience constraint.

This design lets each comparison attribute the observed gain to a specific component of the MD-CDBG rather than to the framework as a whole.

D. Metrics

We report three metrics. The first two are the discounted total utilities accumulated over the horizon, where the discount factor $\gamma \in (0, 1)$ weights early time steps more heavily, reflecting the greater value of preventing early compromise:

$$U^D = \sum_{t=0}^{T-1} \gamma^t U^D(\sigma^{D,t}, \mu^t), \quad U^A = \sum_{t=0}^{T-1} \gamma^t U^A(\sigma^{A,t} | \theta_A^*).$$
(32)

A higher U^D is better for the defender; for U^A , a value closer to zero indicates a more rational, less self-defeating attacker model (a strongly negative U^A means the attacker is forced into moves that harm itself, which signals an unrealistic adversary model rather than a defender success). The third metric is the violation rate

$$VC = \frac{1}{T} \sum_{t=0}^{T-1} \mathbf{1}[\Omega(t) < \Omega_{\min}],$$
(33)

the fraction of time steps during which the resilience constraint is breached. A lower VC means the system maintains service continuity more consistently and is the primary indicator of the defender’s true objective. When comparing two methods we report the relative violation reduction

$$\Delta_{VC} = (VC_{\text{base}} - VC_{\text{ours}}) / VC_{\text{base}}$$

i.e. the fractional decrease in violation rate of our method with respect to a baseline; all reported metrics are averaged over 10 independent replications.

E. Results

Table III summarizes the four-method comparison, and Figures 3–5 present the same data graphically.

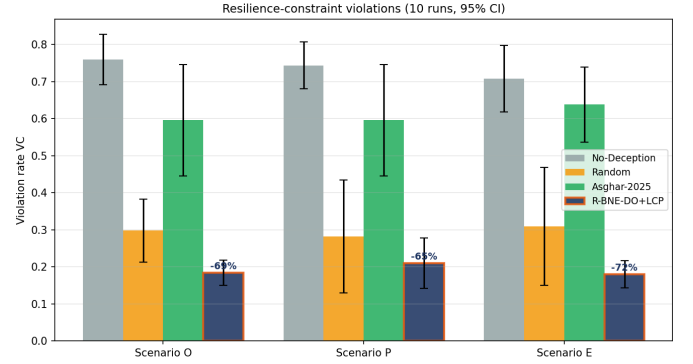


Fig. 4. Resilience-constraint violation rate VC (10 runs, 95% CI). R-BNE-DO+LCP reduces violations by 65–72% relative to Asghar-2025 across all scenarios.

TABLE III
FOUR-METHOD COMPARISON (MEAN OVER 10 RUNS). LOWER VC IS BETTER; LESS NEGATIVE U^A INDICATES A MORE RATIONAL ATTACKER.

Sc	Method	U^D	U^A	VC
O	No-Deception	3.51	−0.36	0.76
	Random	5.47	−0.27	0.30
	Asghar-2025	5.25	−1.77	0.60
	R-BNE-DO+LCP	5.62	−0.68	0.18
P	No-Deception	3.50	−0.35	0.74
	Random	5.24	−0.30	0.28
	Asghar-2025	5.25	−1.77	0.60
	R-BNE-DO+LCP	5.56	−0.63	0.21
E	No-Deception	3.71	−0.32	0.71
	Random	5.26	−0.27	0.31
	Asghar-2025	5.19	−1.78	0.64
	R-BNE-DO+LCP	5.79	−0.56	0.18

Resilience (Fig. 4). This is the headline result. R-BNE-DO+LCP achieves the lowest violation rate in every scenario ($VC = 0.18$ – 0.21). Applying the relative violation reduction Δ_{VC} defined above, this is a 69%, 65%, and 72% decrease relative to Asghar-2025 in scenarios O, P, and E respectively (e.g. $(0.60 - 0.18)/0.60 \approx 0.69$ for O), and up to 76% relative to No-Deception. The full ordering is consistent across scenarios: $R-BNE-DO < Random < Asghar-2025 < No-Deception$. That Random outperforms the more sophisticated Asghar-2025 on resilience is itself revealing: methods that optimize a non-resilience objective can actively steer the system into low-resilience states, whereas random placement at least spreads risk.

Defender utility (Fig. 3a). R-BNE-DO+LCP dominates in scenarios P (5.56 vs 5.25) and E (5.79 vs 5.19), and is statistically indistinguishable from the best baseline in scenario O (overlapping 95% confidence intervals). The interpretation is important: against the weakest attacker (O), aggressive greedy placement and equilibrium placement perform comparably on raw utility, but our method still secures the system far better (VC 0.18 vs 0.60). In other words, the defender obtains equal or higher utility and substantially better resilience, it does not trade one for the other.

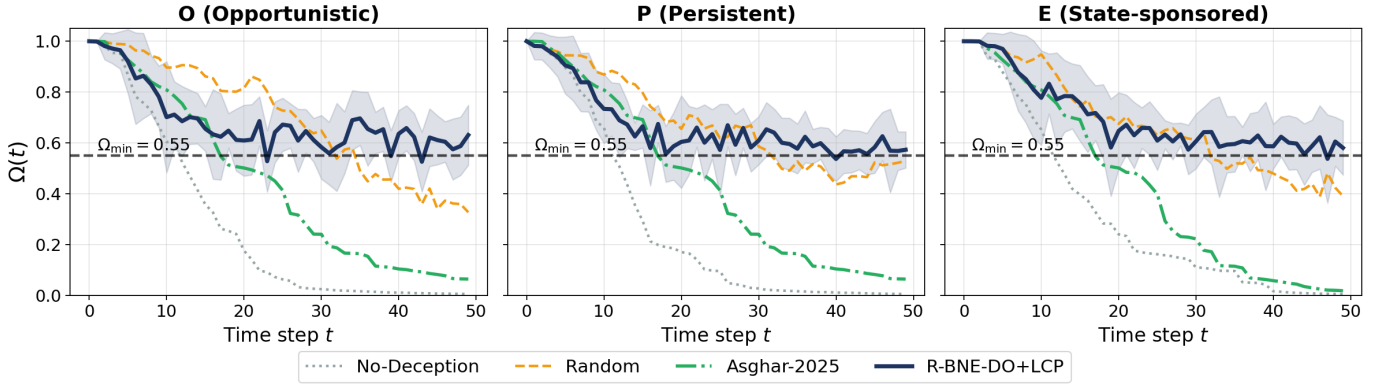


Fig. 5. Global resilience $\Omega(t)$ over time (10 runs, shaded $\pm$ std). R-BNE-DO+LCP maintains $\Omega(t)$ above $\Omega_{\min}$ far more consistently than the baselines.

Attacker rationality (Fig. 3b). The zero-sum proxy of Asghar-2025 drives the attacker utility to ≈ -2.0 , i.e. an adversary systematically forced into self-harming moves. Our non-zero-sum treatment yields $U^A \approx -0.6$, a rational attacker that plays to its genuine incentives. This gap is not a defender win, it signals that the zero-sum model misrepresents the threat, which can lead to defenses tuned against an unrealistic adversary.

Temporal behavior (Fig. 5). The resilience trajectories show that R-BNE-DO+LCP stays above $\Omega_{\min}$ most of the time and rebounds quickly after compromise spikes, a direct effect of the fallback isolation policy triggered by the feasibility check. The baselines, by contrast, drift below the threshold and remain there, never recovering.

F. Discussion

Beyond the per-figure reading above, two higher-level lessons emerge. First, the comparison design (Section VII) lets us attribute the resilience gain to specific causes: the gap to Asghar-2025 isolates the value of the non-zero-sum (LCP) treatment and the resilience constraint, while the gap to Random isolates the value of strategic, equilibrium-based placement over the mere presence of decoys. Both gaps are large, confirming that neither component alone suffices. Second, the consistency of the ordering across three very different attacker profiles suggests the advantage is structural rather than tuned to a particular adversary: embedding resilience as a hard constraint protects service continuity regardless of who is attacking.

VIII. CONCLUSION

We presented the MD-CDBG, a multi-domain cyber-deception Bayesian game that couples the three IoT layers under formal resilience constraints, together with the R-BNE solution concept and the R-BNE-DO algorithm. By correctly modeling the non-zero-sum structure via an LCP oracle and embedding resilience as a hard constraint, the framework reduces resilience violations by 65–72% relative to the state-of-the-art GameSec 2025 approach while inducing rational attacker behavior.

Several directions extend this work; while the present results are established in simulation, the framework is designed to carry over to deployed settings. (i) Learning-based oracle: the greedy and equilibrium oracles could be complemented by a trained deep detector (e.g., a graph attention network or transfer autoencoder) acting as an additional best-response oracle within the double-oracle loop, combining the forward-looking guarantees of game theory with the pattern-recognition strength of machine learning. (ii) Symmetric information: a symmetric extension would model the attacker’s own belief over the defender type, turning the game into a two-sided incomplete-information interaction. (iii) Richer deception techniques: the defender’s action repertoire can be extended with additional network-layer mechanisms such as software-defined network reconfiguration and deceptive routing or address-shuffling, each adding new defender actions while preserving the multi-domain coupling. (iv) Scalability: approximate equilibrium solvers would let the model scale to IoT deployments with thousands of heterogeneous nodes. Together, these directions point toward resilient-by-design cyber-deception, where security and service continuity are optimized jointly rather than traded off against each other.

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